

VENKATRAMANAN MAHENDRARAJAN

Membrane vesicles | Host–Microbe Interactions | Multi-Omics | Biofilms |

Antimicrobial Peptides | Microbiome | Immunology | Quorum Sensing

ICMR Project Scientist & Doctoral Research Fellow

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[Website](#) | [ORCID](#) | [Google Scholar](#) | [Scopus](#) | [Web of Science](#) | [ResearchGate](#) | [LinkedIn](#)

1. Professional Profile

Microbiologist and computational biologist with expertise in bacterial membrane vesicles, extracellular vesicle biology, antimicrobial peptides, molecular microbiology, host–microbe interactions, and multi-omics integration. Experienced in combining experimental microbiology with genomics, proteomics, transcriptomics, molecular docking, molecular dynamics simulations, and preclinical animal models. Author of two published patents, 13 peer-reviewed publications and one preprint.

2. Academic Background

PhD in Microbiology (Thesis Submitted; Viva Voce Pending)

Vellore Institute of Technology, India | 2022–Present

Integrated M.Sc. Life Sciences

Central University of Tamil Nadu, India | 2015–2020

CGPA: 8.3/10 (First Class with Distinction)

3. Professional Experience

ICMR Project Scientist (Indian Council of Medical Research, Government of India)

Vellore Institute of Technology, India | 2024–Present

- Investigating bacterial membrane vesicles, antimicrobial peptides, host–microbe interactions, immunomodulation, proteomics, and translational microbiology.

Doctoral Research Fellow (Vellore Institute of Technology Research Fellowship)

Vellore Institute of Technology, India | 2022–Present

- Investigating probiotic membrane vesicles and their antimicrobial and anti-inflammatory properties through proteomics-guided antimicrobial peptide discovery and computational and experimental validation.

SERB Junior Research Fellow (Science and Engineering Research Board, Government of India)

Central University of Punjab, India | 2021–2022

- Investigated molecular mechanisms underlying antifungal drug resistance in yeast using molecular genetics and functional genomics approaches.

4. Teaching Experience

Teaching and Research Assistant

Vellore Institute of Technology, India | 2022–2024

- Served as an operator of flow cytometry facility for two years.
- Assisted in undergraduate and postgraduate microbiology laboratory instruction.
- Supervised undergraduate and postgraduate student research projects.
- Conducted laboratory demonstrations, assessments, and examination invigilation.

5. Research Interests

- Bacterial Membrane Vesicles (MVs/OMVs)
- Host–Microbe Interactions
- Antimicrobial Peptides
- Molecular Microbiology
- Immunology and Inflammation
- Bioinformatics and Systems Biology
- Molecular Dynamics Simulations

6. Research Expertise

Membrane Vesicle Isolation & Purification

Ultracentrifugation, Ultrafiltration, Tangential Flow Filtration, Affinity-based purification

Vesicle Characterization

Dynamic Light Scattering, Zeta Potential Analysis, Nanoparticle Tracking Analysis, TEM, FESEM, SDS-PAGE, Flow Cytometry, Proteomics, Vesicle Transcriptomics

Microbiology

Aerobic/anaerobic bacterial culture, yeast culture, antimicrobial susceptibility testing, MIC determination, quorum sensing assays, biofilm assays including SEM & CLSM, probiotic characterization, hydroxyapatite enamel model

Molecular Biology

DNA, RNA and protein extraction, PCR, RT-qPCR, molecular cloning, bacterial transformation, homologous recombination, PCR validation, Western blotting

Genomics & Sequencing

Nanopore Library preparation, 16S rRNA sequencing analysis, Illumina and Oxford Nanopore sequencing analysis, RNA-seq, Differential expression analysis (edgeR & DESeq2)

Immunology & Cell Biology

Mammalian cell culture (RAW264.7, 3T3, L929), Cell viability assays (MTT), Cytokine profiling, Nitric oxide quantification (Griess assay), ROS analysis, NF-κB signaling studies

Animal Studies

Rodent handling & oral administration, Oral wound and tooth infection models, LPS-induced paw edema model, Repeated-dose oral toxicity (OECD 407), Serum biochemistry, Tissue collection, Histopathology (paraffin embedding, microtomy, H&E staining)

Computational Biology & Bioinformatics

Proteomics data analysis, Genome annotation, Comparative genomics, Transcriptomic analysis, Molecular docking, Molecular dynamics simulations, Antimicrobial peptide screening, Statistical analysis

Computational Tools & Software

Bioinformatics: ThermoRawFileParser, OpenMS, MS-GF+, MaxQuant, PeptideShaker

Structural Biology: AutoDock4, AutoDock Vina, GROMACS, NAMD, CHARMM-GUI, PyMOL, BIOVIA Discovery Studio Visualizer

Omics Analysis: FASTQC, Fastp, AdapterRemoval, Trimmomatic, SPAdes, dnaapler, RAST, DFAST, CheckM, Proksee, BWA-MEM, SAMtools, edgeR, DESeq2, MetaPhlAn, HUMAnN

Statistical & Image Analysis: GraphPad Prism, Fiji (ImageJ)

Programming & Computing: R (Biostatistics, RNA-seq analysis), Python (Bioinformatics workflows), Linux

Scientific Visualization & Productivity: BioRender, Microsoft Office, LibreOffice

7. Major Scientific Contributions

Membrane Vesicle Therapeutics

Demonstrated antimicrobial and antibiofilm activity of *Lactobacillus acidophilus* membrane vesicles against *Streptococcus mutans* using in silico, in vitro and in vivo infection models.

Antimicrobial Peptide Discovery

Integrated LC-MS/MS proteomics, computational screening, molecular dynamics simulations, peptide synthesis, and biological validation to identify membrane-active antimicrobial peptides derived from probiotic vesicles.

Anti-Inflammatory Vesicle Research

Established anti-inflammatory effects of probiotic membrane vesicles through suppression of NF- κ B signaling and reduction of inflammatory mediators in cellular and animal models.

OMV Isolation Technology

Developed a process for extraction and purification of OMVs from complex biofilm architectures, enabling enhanced recovery and characterization of biofilm-associated vesicles (Applied for patent).

8. Publications and Research Metrics

Peer-reviewed Publications: 13

First-author publications: 12

Preprint: 1

Citations: 244

h-index: 6

i10-index: 5

9. Intellectual Property (Patent Published)

[1] Easwaran N, **Mahendrarajan V**, Lazarus HPS.

Process for Extraction and Purification of Outer Membrane Vesicles from Complex Biofilm Architectures of *Chromobacterium violaceum*

Application No. 202541131837 A.

[2] Easwaran N, **Mahendrarajan V**.

A Novel Peptide VMNE244 from *Lactobacillus acidophilus* Membrane Vesicles: Synthesis and Antibacterial Activity Against *Streptococcus mutans*

Application No. 202641073848 A.

10. Selected Peer-Reviewed Publications

[1] **Venkatramanan M**, Sankar Ganesh P, Senthil R, Akshay J, Veera Ravi A, Langeswaran K, et al. Inhibition of Quorum Sensing and Biofilm Formation in *Chromobacterium violaceum* by Fruit Extracts of *Passiflora edulis*. *ACS Omega* 2020;5:25605–16. <https://doi.org/10.1021/acsomega.0c02483>. (Q2, IF: 5.2)

[2] **Mahendrarajan V**, Bari VK. A critical role of farnesol in the modulation of Amphotericin B and Aureobasidin A antifungal drug susceptibility. *Mycology* 2022;13:305–17. <https://doi.org/10.1080/21501203.2022.2138599>. (Q1, IF: 5.7)

[3] **Venkatramanan M**, Nalini E. Regulation of virulence in *Chromobacterium violaceum* and strategies to combat it. *Front Microbiol* 2024;15:1–13. <https://doi.org/10.3389/fmicb.2024.1303595>. (Q1, IF: 5.8)

[4] **Mahendrarajan V**, Lazarus H, Easwaran N. Quorum quenching mediated biofilm impediment in *Chromobacterium violaceum* and *Staphylococcus aureus* by leaf extracts of *Delonix elata*. *Heliyon* 2024;10:e31898. <https://doi.org/10.1016/j.heliyon.2024.e31898>. (Q1, IF: 4)

[5] **Mahendrarajan V**, Easwaran N. Quorum quenching effects of linoleic and stearic acids on outer membrane vesicle-mediated virulence in *Chromobacterium violaceum*. *Biofouling* 2024;41:92–102. <https://doi.org/10.1080/08927014.2024.2446930>. (Q2, IF: 2.6)

[6] **Mahendrarajan V**, Nalini E. In silico analysis of bacteriocins from *Lactobacillus acidophilus* membrane vesicles against *Streptococcus mutans* GtfB protein. *J Biomol Struct Dyn* 2025;43:10854–10864. <https://doi.org/10.1080/07391102.2025.2460743>. (Q2, IF: 2.7)

[7] **Mahendrarajan V**, Kodiveri Muthukaliannan, G, Easwaran, N. Quorum quenching mediated virulence factor impairment by *Pandanus tectorius* leaf extract against *Chromobacterium violaceum*. *Natural Product Research* 2025;15:1–8. <https://doi.org/10.1080/14786419.2025.2491837>. (Q1, IF: 2)

- [8] **Venkatramanan M**, Easwaran, N. Isolation, Probiotic Characterization and Genomic Analysis of *Enterococcus durans* VIT3 from Edible Curd. *Microbial Pathogenesis* 2025;205:107649. <https://doi.org/10.1016/j.micpath.2025.107649>. (Q2, IF: 3.9)
- [9] **Mahendrarajan V**, Lazarus HPS, Muthukaliannan GK, Varghese S, Easwaran N. Membrane vesicles from Red Complex bacteria: key players in oral pathogenesis, immune disruption, systemic diseases, and therapeutic insights. *Front Oral Health* 2025;28:1607931. <https://doi.org/10.3389/froh.2025.1607931>. (Q1, IF: 3.4)
- [10] **Mahendrarajan V**, Easwaran N. Integrated proteomics, molecular dynamics, and in vitro characterization of antimicrobial peptide from *Lactobacillus acidophilus* vesicles against *Streptococcus mutans*. *RSC Adv* 2026;16:5743–5757. <https://doi.org/10.1039/D5RA09511E>. (Q2, IF: 6.1)
- [11] Lazarus HPS, **Mahendrarajan V**, & Easwaran N. Genomic insights into *Exiguobacterium indicum* VIT-2023: a dual-function PGPR with potential for plant health protection and plastic biodegradation. *Environmental Pollutants and Bioavailability* 2026;38. <https://doi.org/10.1080/26395940.2026.2636356>. (Q2, IF: 4)
- [12] **Mahendrarajan V**, Easwaran N. Biocompatible membrane vesicles from *Lactobacillus acidophilus* MTCC 10307 exhibit potent anti-inflammatory activity. *bioRxiv* (Preprint). <https://doi.org/10.64898/2026.04.01.715785>
- [13] **Mahendrarajan V**, Easwaran N. *Lactobacillus acidophilus* membrane vesicles demonstrate antimicrobial and antibiofilm activity against *Streptococcus mutans* – an in vitro and in vivo study. *Artificial Cells, Nanomedicine and Biotechnology* 2026;54:17–34. <https://doi.org/10.1080/21691401.2026.2662204>. (Q1, IF: 5.6)
- [14] **Mahendrarajan V**, Sankaranarayanan G, Muthukaliannan GK. et al. Triphala Modulates Membrane Vesicle Transcriptome of *Enterococcus durans* VIT3 to Influence Antibiotic Response and Probiotic Functions. *Molecular Biotechnology* 2026. <https://doi.org/10.1007/s12033-026-01578-x> (Q3, IF: 3.2)

11. Professional Memberships

- American Society for Microbiology (ASM) – Contributing Member
- Indian Extracellular Vesicles Society (EXO-RISING)

12. Research Trainings

- Extracellular Vesicles: Isolation, Characterization & Applications, AMRF, Madurai (2024).
- Training on the state of art advanced multi omics techniques in Molecular Oncology, ICMR, Govt. of India, Cancer Institute (WIA), Chennai (2025).

13. Achievements

- GATE 2020 (Life Sciences) – Qualified (78.6 percentile).
- Awarded SERB Junior Research Fellowship.
- Appointed as ICMR Project Scientist.

14. Peer Review Activities

Reviewer for *BMC Microbiology* (ISSN: 1471-2180)

Reviewer for *Folia Microbiologica* (ISSN: 1874-9356)

15. Ongoing Research Work

Parkinson's Disease Multi-Omics Integration

Conducting integrative multi-omics analysis combining brain transcriptomics and gut microbiome metagenomics datasets from control and Parkinson's disease cohorts to investigate the gut–brain axis. Performed differential gene expression analysis using DESeq2 and carried out comprehensive microbiome profiling using HUMAnN and MetaPhlAn. Mapped host–microbiome interactions to identify functionally relevant associations contributing to neuroinflammation. Applied machine learning approaches, including Random Forest-based feature selection, to identify key predictive signatures and potential biomarkers associated with Parkinson's disease progression

Oral Microbiome Metagenomics Analysis in Wistar Rats

Conducting metagenomic analysis of oral samples from Wistar albino rats treated with Lactobacillus acidophilus, chlorhexidine, and Triphala. Utilizing Illumina sequencing and bioinformatics analysis to investigate alterations in microbial community structure, taxonomic composition, alpha and beta diversity, and differentially abundant taxa, with the goal of understanding treatment-associated modulation of the oral microbiome.

16. Scientific Presentations

2025 - Poster Presentation, IMMUNOCON 2025 (Annual Conference of the Indian Immunology Society), Sher-i-Kashmir Institute of Medical Sciences (SKIMS), Srinagar, India.

2025 - Poster Presentation, International Conference on Contemporary Antimicrobial Research (ICCAR 2025), North-Eastern Hill University (NEHU), Shillong, India.

2024 - Poster Presentation, International Conference on Biotechnology for Environment & Health (ICBEH 2024), Vellore Institute of Technology (VIT), Vellore, India.

2022 - Poster Presentation, IMMUNOCON 2023 (Annual Conference of the Indian Immunology Society), Postgraduate Institute of Medical Education and Research (PGIMER), Chandigarh, India.

17. References

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2. Dr. K. M. Gothandam – Professor, Vellore Institute of Technology

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